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$$\begin{aligned} & 1 + \frac{a-bi}{a+bi} + \frac{a+bi}{a-bi} \\ &= 1 + \frac{(a-bi)(a-bi)}{(a+bi)(a-bi)} + \frac{(a+bi)(a+bi)}{(a-bi)(a+bi)} \\ &= 1 + \frac{a^2 - 2abi - b^2}{a^2 + b^2} + \frac{a^2 + 2abi - b^2}{a^2 + b^2} \\ &= \frac{3a^2 - b^2}{a^2 + b^2} \end{aligned}$$

References